

SNF-Bench: Separating Static Drift from Natural Flow in Long-Horizon Fixed-Camera Video Generation

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Abstract

Long-horizon video generation is evaluated with whole-frame metrics that reward motion and temporal consistency. For fixed-camera nature scenes this creates an ambiguity: motion of water, fire, smoke, or rain is desirable, whereas motion of the background is an error. A system can therefore score well on motion while its scene drifts, or on consistency while its flow stagnates. We introduce SNF-Bench, an evaluation framework for long-horizon fixed-camera generation that partitions each scene into static support and dynamic flow and reports static fidelity, flow persistence with absolute magnitude, and drift leakage separately and never as one score, the last as an interpretive diagnostic rather than a headline measurement. Each factor is validated mechanistically rather than by correlation with preference: we inject global translation, rotation, and scale drift and progressive motion attenuation at known severity into real generations, and require each factor to respond in its stated direction and to remain selective against corruptions it does not target. Auditing publicly released long-horizon text-conditioned checkpoints under one common recorded inference configuration, plus an image-conditioned deployment-sensitivity panel reported separately, we find that whole-frame motion and static-region drift induce near-opposite orderings of the same outputs, and that controlled drift which triples our static-fidelity factors moves a widely used whole-frame motion score by roughly a tenth—in the direction that rewards it. SNF-Bench measures where motion occurs and whether it persists; it does not measure physical realism.

1. Introduction

Autoregressive (AR) and streaming video diffusion models now extend generation from seconds to minute-scale horizons [12, 22, 23]. Evaluation has not kept

pace: small temporal errors invisible in short clips accumulate over long rollouts into background displacement, color drift, or motion decay, and whole-frame metrics summarize these qualitatively different failures into one score.

The problem is sharpest in fixed-camera nature scenes. In a static-camera video of a river, the water should move while the bank, rocks, and surrounding landscape stay fixed. The same separation governs waterfalls, ocean waves, rainfall, fire, smoke, and wind-driven vegetation: some regions form static support that must remain structurally fixed, while others carry dynamic flow that must persist. Whole-frame motion measures reward optical flow regardless of its origin, crediting background drift as readily as genuine fluid motion, while temporal-consistency measures can favor videos whose stability comes from suppressing the flow they should sustain. Two opposite failures—static drift and flow decay—are thereby conflated, and increasingly so at long horizons, where accumulated background motion can dominate a sequence’s apparent dynamics. Figure 1 shows both failures in outputs of released checkpoints under the common setting.

We introduce SNF-Bench (Static-Natural Flow), an evaluation framework built around this decomposition, evaluated at 60 and 120s horizons with shorter (5s) and longer (240s) tiers retained as diagnostics. Measurement is organized into three factor groups, the third interpretive. Static fidelity captures geometric displacement and residual flow inside nominally stationary regions. Flow persistence reports the retention of dynamic-region motion with its absolute compensated magnitude, so a ratio cannot look strong merely because motion is uniformly weak. Drift leakage relates static-region motion and global-motion compensation to dynamic-region flow; it contextualizes the two measurement axes rather than adding a third, and the central claim does not rest on it.

A benchmark is only as trustworthy as its measurements, so we validate the protocol independently of

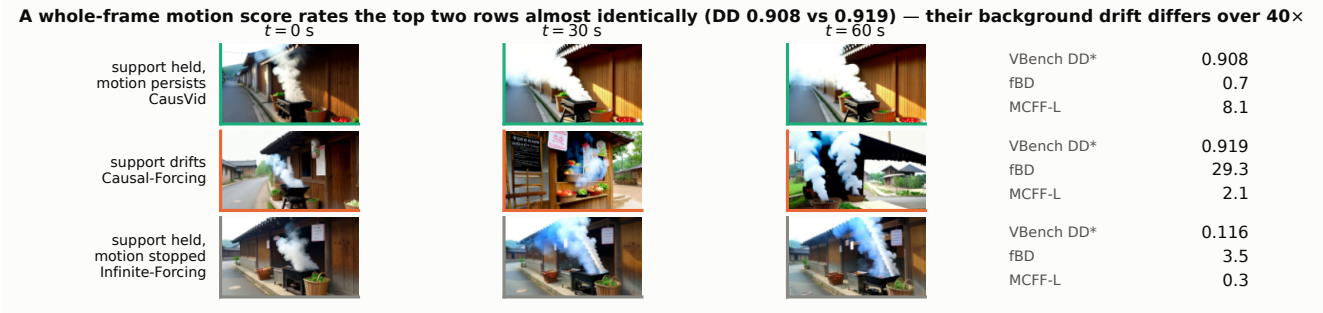


Figure 1. Why a fixed-camera benchmark needs a spatial decomposition. Three 60s outputs from released checkpoints under the common matched setting. A whole-frame motion score rates the first two rows almost identically, yet only the first holds its scene in place; their static-region drift differs by over an order of magnitude. The third row shows the opposite failure, which that score does detect. DD* is the per-video quantity underlying VBench [8] Dynamic Degree, computed under its published rule but read before the binarization the suite applies when aggregating, so that near-identical rows remain distinguishable as numbers.

any model comparison. Controlled perturbations inject global translation, rotation, and scale drift, progressive motion attenuation, late freezing, photometric drift, and temporal repetition into real fixed-camera generations at known severity, and we require each factor to respond in the direction its definition states and to remain selective against corruptions it does not target. Because a parameter expressed in degrees is not commensurate with one expressed in pixels, geometric severity is specified as the mean displacement a corruption induces, which makes the families directly comparable. We additionally recompute every measurement under systematic erosion and dilation of the region partition.

We then audit publicly released long-horizon generators as executed through the recorded evaluation pipeline. The audit exposes cases where conventional metrics and SNF-Bench interpret the same outputs differently: high whole-frame motion coexisting with substantial static-region displacement, and high stability coexisting with decaying flow. We do not argue that existing metrics are incorrect. They answer a whole-frame question accurately; that question does not encode the spatial distinction this setting requires. Nor does SNF-Bench measure physical realism—a sequence may be drift-free and persistent yet physically implausible. The contribution is narrower and complementary: making the spatial origin and temporal persistence of motion explicit.

Our contributions are:

1. SNF-Bench, an evaluation framework for long-horizon fixed-camera generation organized around a static-support/dynamic-flow partition.
2. A measurement protocol that reports static fidelity, flow persistence with absolute magnitude, and drift

leakage separately, never as a single score, the last as an interpretive diagnostic.

3. Mechanistic validation of every reported factor under corruptions of known type and severity, with selectivity measured against the families each factor should ignore and every measurement recomputed under erosion and dilation of the region partition.
4. A public-model audit showing that this decomposition materially changes the interpretation and ordering of long-horizon behavior relative to whole-frame metrics.

2. Related Work

Video-generation evaluation. General-purpose suites assess visual quality, temporal consistency, motion, and semantic alignment. VBench [8] decomposes quality into dimensions including background consistency, motion smoothness and dynamic degree, and VBench-2.0 [26] extends it toward intrinsic faithfulness; Eval-Crafter [13] learns a weighting of many metrics from user opinion; Video-Bench [4] scores generations with multimodal language models validated against human preference. Others specialize: T2V-CompBench [19] targets compositional binding, VMBench [11] perceptually aligned motion quality, and WorldScore [1] controllability and dynamics for world generation. These set the standards a benchmark paper must meet—explicit taxonomy, validated metrics, stated scale—but their measurements are whole-frame or sequence-level, and such an aggregate cannot reveal whether motion originates where intended or in displacement of nominally static structure. VMBench is the closest comparator, but it asks whether motion is perceptually plausible where we ask where it occurs and whether it survives compensation for global displacement. Our

validation is correspondingly mechanistic—perturb by a known amount, require the expected response—complementing the human-alignment evidence those suites provide.

Long-horizon autoregressive generation. Autoregressive and streaming generators extend synthesis beyond short bidirectional clips by producing frames or temporal chunks causally, including CausVid [24], Self-Forcing [7], Infinite-Forcing [9], Rolling-Forcing [12], Reward-Forcing [14], LongLive [22], and Causal-Forcing [27] with its few-step successor Causal-Forcing++ [25]; bidirectional systems such as LTX-Video [3] and Wan [21] provide short-horizon reference points. These systems differ substantially in temporal block size, training objective, memory and cache policy, and horizon-extension mechanism [23], yet their evaluation typically relies on generic video metrics or method-specific qualitative analysis. That diversity motivates a controlled common-setting audit, and the evaluation gap motivates a system-agnostic question: does static support remain fixed while dynamic flow persists?

Image dynamics and nature-flow animation. Work on animating still images has long modeled natural motion explicitly: controllable fluid animation separates static image structure from specified fluid motion [15], and Generative Image Dynamics models long-term scene motion through learned trajectory representations [10]. These works treat static support and dynamic content as distinct modeling targets; SNF-Bench elevates that distinction to an evaluation principle for long-horizon generators.

3. SNF-Bench Design

SNF-Bench evaluates long-horizon generation under a deliberately constrained setting: a fixed camera observing a scene containing both stationary structure and persistent natural motion. This isolates two failure modes that are difficult to separate in unconstrained video—static drift, where nominally stationary structure moves over time, and flow decay, where intended dynamics weaken or disappear. Every sequence is treated as a composition of two spatial roles, Ω_{static} for regions expected to remain geometrically stationary and Ω_{flow} for regions expected to exhibit sustained motion. The partition defines an evaluation role, not a semantic ontology: a riverbank is static support while the river surface is flow; buildings are static support in a rainfall scene while the falling rain is flow. SNF-Bench does not require dynamics inside Ω_{flow} to follow

any physical model. It measures whether motion stays localized to the appropriate regions and persists over time without mixing static support. Every system receives identical conditioning per item, and no prompt is rewritten for a particular method. Both conditioning modalities are evaluated under the same decomposition, differing only in how static support is established: the text-conditioned setting builds it in the opening of the sequence, while the image-conditioned setting takes it from a shared source image, so divergence there is attributable to the system rather than to a differently imagined scene. Both tracks are reported here at the 60 and 120 s horizons, along with 5 and 240 s in the supplementary.

3.1. Scene Categories and Horizons

Prompts are stratified across representative fixed-camera natural processes—channel water, ocean waves, precipitation, fire and smoke, lava, and wind-borne motion—with the category set fixed before the final sweep. The realized composition is dominated by directional and broad-area water motion with additional atmospheric and combustion categories, and conclusions should be read at that scope. The strata are not equally populated: channel water accounts for 61% of the principal prompt set. Reported aggregates are means over prompts, not category macro-averages: three of the five populated categories hold two prompts each, so equal category weighting would buy balance at a cost in variance the sample cannot support. We instead recompute every system both ways and state which conclusions survive, with complete per-category counts in the supplementary.

Four temporal regimes are used: 5 s as an initialization check, 60 s as the primary reported horizon, 120 s as a supporting tier, and 240 s as a diagnostic extreme. Conclusions rest on the 60 s tier; the 120 s tier is reported alongside it as a supporting check, and the 5 and 240 s tiers are in the supplementary material. A run that cannot reach a target horizon under its declared track and setting is reported as unsupported rather than silently omitted.

3.2. Region Partition

The partition is obtained automatically and identically for every sequence. Mean optical-flow magnitude over the leading 12% of sampled frame pairs is thresholded by Otsu’s method [16] into a dynamic region and its complement; the static region is then eroded and a 4% frame border discarded, which together form an ignored transition band so that boundary pixels—where the partition is least certain and flow least reliable—contribute to neither region. Restricting the estimate

to an early window reduces circularity from late accumulated drift: at 60s the partition is fixed from the first 7.1s and is therefore not defined by the late-window failure being measured. It still depends on early checkpoint behavior and on the same flow estimator used downstream. Two limitations bound the later conclusions. The partition is automatic, so a system drifting within the first 12% of its rollout can influence its own partition; we therefore treat partitions as measured quantities and recompute every result under systematic erosion and dilation of the boundary (Sec. 5). And a two-way partition cannot represent layered content: rain and snow transit static support, so those pixels are assigned by flow magnitude alone and static-fidelity values in precipitation scenes carry an unmodeled contribution. Derivation, the sensitivity study, and the annotation protocol we regard as the natural strengthening of this component are in the supplementary material.

3.3. Benchmark Manifest

SNF-Bench is defined by a versioned manifest fixed before the final sweep, recording per item: identifier and category, exact prompt, target horizons, resolution and frame-rate metadata, and provenance. After freezing, items are never removed because a system performs poorly; corrections require a new benchmark version. A failure taxonomy—unsupported horizon, out-of-memory, runtime crash, numerical corruption, invalid output—is likewise fixed in advance, and failures are reported alongside successful generations rather than silently excluded. Coverage is counted from valid per-video measurements rather than from the presence of a results file, so a run that terminated early is reported as partial.

4. Drift–Motion Evaluation Protocol

SNF-Bench evaluates fixed-camera generation along three factor groups: Static Fidelity and Flow Persistence are the measurement axes, and Drift Leakage (DLR, DAR) contextualizes them rather than adding a third measurement. No individual score is sufficient: low background motion can be achieved by freezing the entire video, and high foreground motion can be produced by drifting the entire scene. We therefore always report static-region stability jointly with both the relative persistence and absolute magnitude of dynamic-region motion. Let I_t denote frame t , Ω_{static} and Ω_{flow} the region masks of Sec. 3.2, and $\mathbf{u}_t$ the optical flow from I_t to I_{t+1} , estimated with RAFT [20]. Each sequence defines an early window $\mathcal{E}$ and a late window $\mathcal{L}$ of equal duration, $\mathcal{E}$ beginning after a warm-up so that generation transients do not define the motion refer-

ence; extents are fixed in the specification.

4.1. Static Fidelity

Feature-aligned Background Drift (fBD). ORB features [18] extracted inside Ω_{static} from the first frame are matched against each late-window frame under robust geometric filtering; fBD ($\downarrow$) is the median matched-point displacement, averaged over the late window and expressed as a percentage of the frame diagonal. It captures accumulated geometric displacement of persistent structure and is intentionally insensitive to fine-scale flicker that moves no structure (formal definition in Supp. S1).

Normalized Background Flow (NBF). Complementing accumulated drift, NBF ($\downarrow$) is the mean optical-flow magnitude inside the static mask, averaged over all frame pairs and normalized by frame width for resolution invariance and by the inter-frame interval Δt for frame-rate invariance, giving units of 10^{-3} frame widths per second (Supp. S1). Here Δt is the interval of the sampled pairs actually used for flow estimation: methods are evaluated at their native frame rates, which differ across the roster, and the evaluation pipeline subsamples each clip to a common temporal rate before measurement. Expressing NBF per unit time rather than per frame makes the quantity independent of both the native rate and the sampling policy, so the measure remains comparable if either changes. NBF is a normalized magnitude, not a ratio: dividing by whole-frame flow would couple background leakage to foreground behavior and inflate the measure for methods whose dynamic motion decays.

4.2. Flow Persistence

Global-motion compensation. Coherent scene displacement is removed before dynamic-region motion is evaluated. A single translation vector is insufficient: under rotation or zoom the static-region flow field is not constant, and its component-wise median is near zero for a centered rotation, so a translation-only estimate would leave exactly the corruptions Sec. 5 injects uncompensated. We therefore estimate a robust global similarity transform $T_t(\mathbf{x}) = s_t R_t \mathbf{x} + \mathbf{b}_t$ from static-region correspondences under RANSAC, and subtract the displacement field it induces from the measured flow pointwise, giving a compensated field $\tilde{\mathbf{u}}_t$ (Supp. S1). A similarity rather than an unrestricted affine model is used because affine shear absorbs legitimate local deformation and over-corrects. Where correspondences are insufficient we fall back to median translation and record the fallback per sequence.

345 Motion-Compensated Foreground Flow (MCFF).
 346 MCFF ($\uparrow$) is the mean compensated flow magnitude
 347 inside Ω_{flow} over a temporal window. We report it on
 348 both the early and the late window—MCFF-E and
 349 MCFF-L—so a method with uniformly negligible motion
 350 cannot appear strong through relative measures
 351 alone (Supp. S1).

352 Flow Persistence (FP). FP ($\uparrow$) is the ratio of late to
 353 early MCFF, clipped at a fixed constant to prevent
 354 unstable ratios when early motion is small (Supp. S1).
 355 FP near one indicates retained motion; substantially
 356 smaller values indicate decay; values above one are not
 357 automatically better. FP is never reported without its
 358 MCFF magnitudes.

359 4.3. Drift Leakage and Attenuation

360 Whole-frame metrics cannot determine whether appar-
 361 ent dynamic-region motion arises from intended flow or
 362 from displacement of the whole scene; we report two di-
 363 agnostics, since neither subsumes the other. Let F_{static}
 364 denote mean flow magnitude over Ω_{static} , and F_{raw} ,
 365 F_{comp} the mean raw and compensated magnitude over
 366 Ω_{flow} , all in the late window, where drift has accumu-
 367 lated.

368 Drift Leakage Ratio (DLR). DLR ($\downarrow$), the ratio of
 369 static-region to raw dynamic-region flow magnitude in
 370 the late window, locates the sequence’s motion energy
 371 (Supp. S1). It is not a fraction and is deliberately
 372 unbounded: $\text{DLR} > 1$ indicates that static support
 373 moves more than the region intended to move, a regime
 374 that occurs in practice.

375 Drift Attenuation Ratio (DAR). DAR ($\downarrow$), one minus
 376 the ratio of compensated to raw dynamic-region flow,
 377 measures the signed relative change in dynamic-region
 378 motion after compensation—how much apparent motion
 379 a global displacement model accounts for (Supp.
 380 S1). Reported values are clipped to $[0, 1]$ while the
 381 signed quantity is retained for validation; negative val-
 382 ues, which arise where local flow opposes the estimated
 383 global field, are reported rather than suppressed.

384 Interpretation. Neither is a causal decomposition of
 385 motion into drift and non-drift components. DLR and
 386 DAR are secondary diagnostics that divide the work
 387 between them: DLR is a translational-leakage diag-
 388 nostic whose rotational response is inverted, so rota-
 389 tional drift is carried by fBD and DAR instead, and
 390 that division is why DLR is never read alone. DAR
 391 is signed, and its negative values are real. High DLR
 392 with moderate DAR indicates static-region motion a

rigid model does not account for; it does not identify
 what the residual is. 393 394

Semantic guardrail. The decomposition is agnostic
 to what moves: a sequence rendering drifting fog
 where a river was requested can occupy a favor-
 able operating point. The natural guardrail is a
 prompt-alignment score—CLIP-based image-text sim-
 ilarity [6], or a learned video-quality predictor [5] at
 higher cost. We do not report one, and the factors
 should be read accordingly: they establish where mo-
 tion occurs and whether it persists, not that the de-
 picted phenomenon is the one requested. Pairing SNF-
 Bench with a semantic score is how it should be used
 in practice. 395 396 397 398 399 400 401 402 403 404 405 406

407 4.4. Joint Interpretation

No factor is read alone: low drift with low compensated
 motion indicates an over-stabilized sequence, high com-
 pensated motion with high static-region flow indicates
 motion contaminated by displacement, and high reten-
 tion with negligible magnitude indicates persistent but
 trivial motion. SNF-Bench therefore reports no single
 scalar score, since collapsing the factors would recreate
 the ambiguity it exists to expose. 408 409 410 411 412 413 414 415

Implementation convention. Videos are normalized to
 a common evaluation resolution and subsampled to a
 common flow-estimation rate; no frame interpolation
 is performed, since synthesizing frames would alter the
 flow field being measured. Every estimator, threshold,
 window extent and constant is fixed globally before the
 sweep and applied unchanged to every system, and the
 per-sequence frame rate, resolution and compensation
 mode are released alongside the scores. 416 417 418 419 420 421 422 423 424

425 5. Controlled Validation

Before SNF-Bench is used to compare generators, we
 validate that its measurements respond to the failure
 modes they target. A benchmark is not justified merely
 because its metrics rank differently; each must change
 selectively, monotonically, and in the expected direc-
 tion under known corruptions. We therefore construct
 controlled variants of fixed-camera reference clips with
 known corruption type and severity, independently of
 the audit in Sec. 6: perturbation strengths, masks, win-
 dows, and hyperparameters are fixed before the final
 results are inspected. 426 427 428 429 430 431 432 433 434 435 436

437 5.1. Perturbations and Criteria

Given a reference sequence we generate variants
 that isolate individual components of the decom- 438 439

position. Geometric drift applies a temporally increasing global transformation—translation, rotation, or scale—simulating camera-like background displacement that accumulates as long-horizon error does. Motion attenuation progressively blends dynamic-region content toward an early reference, and late freezing replaces all content after a chosen fraction of the sequence, both leaving static support untouched. Photometric drift varies brightness and color temperature while preserving geometry and motion, bounding the residual sensitivity of flow-based quantities. Temporal repetition replaces late dynamic content with a short repeated cycle. Partition perturbation erodes and dilates the region boundary rather than the video.

Geometric severity is specified as the mean pixel displacement a corruption induces, with each family’s parameter derived from it using the frame geometry. This is not presentational: a parameter in degrees is not commensurate with one in pixels, and at the evaluation resolution one degree of rotation induces only a few pixels of displacement, so families compared by their native parameters differ in injected severity by more than an order of magnitude. A factor is admitted only if its response direction matches its definition, that response is monotonic in severity, unrelated corruptions produce substantially smaller changes, and conclusions survive perturbation of the region partition. Factors failing these checks are reported as diagnostics rather than headline quantities. The severity specification and the admission outcome for every factor are in the supplementary material.

Complementary checks. Mechanistic validation shows a factor responds to the corruption it names, not that its ordering matches what an observer perceives, and we do not close that gap. The supplementary material specifies an axis-specific protocol—forced choice on one failure axis, never on which output is better—and reports the one instance we ran: an open-weight judge [17] failed a presentation-order control, so no agreement rate is drawn from it. Perceptual agreement remains open.

5.2. Validation Results

Figure 2 and Table 1 report the outcome. Injected translation produces the expected monotone increase in both static-fidelity factors, and progressive late freezing drives MCFF-L and FP down—to 0.02 and 0.07 of their unperturbed values once the late window is frozen—while leaving the static factors comparatively unchanged, which is the selectivity the decomposition requires. The comparison that motivates the benchmark appears in the same experiment. Injected dis-

Factor	target	ρ	ρ_{rot}	off-target	admitted
fBD	drift	+1.00	+0.70	14%	✓
NBF	drift	+1.00	+0.60	6%	✓
MCFF-L	freeze	+1.00	n/a	14%	✓
FP	freeze	+0.80	n/a	21%	✓
DLR	translational leakage	+1.00	-0.80	5%	✓
DAR	drift	+0.90	+1.00	18%	✓
VBench DD	drift	+0.70	+0.70	2%	—

Table 1. Mechanistic validation and factor admission. Each factor is scored on the corruption its definition says it should detect: accumulating global drift for the static-fidelity and drift diagnostics, progressive freezing for the persistence factors. ρ is Spearman correlation against injected severity, signed so that a correct response is positive; ρ_{rot} repeats it for rotational drift, the harder case. off-target is the largest relative excursion under corruptions the factor should ignore—photometric drift and perturbation of the region partition—so a small value indicates selectivity. Admission requires $\rho \geq 0.7$ and off-target response below 25% on the factor’s target family. Rotational response is reported but not gated, and the factors divide that work explicitly: DLR is a translational-leakage diagnostic whose rotational response is inverted (−0.80), while rotational drift is carried by fBD and DAR at +0.70 and +1.00. That division is why DLR is never read alone. VBench DD is a comparison row rather than a candidate factor—it is put through the identical corruptions to show what a whole-frame score does under them—so its admission cell is left blank. Twelve reference clips per severity level.

placement that carries fBD to $3.2\times$ its unperturbed value moves VBench Dynamic Degree [8] to only $1.1\times$, and in the wrong direction: a whole-frame motion score is largely insensitive to background drift, and to the extent it responds at all it improves on precisely the corruption a fixed-camera evaluation should penalize.

Two results qualify the framework rather than support it. Rotation is the harder geometric case for every factor, since correspondences degrade and the induced displacement is smallest at the frame center: both static-fidelity factors respond more weakly than under translation (+0.70 for fBD, +0.60 for NBF, against +1.00 for both), so neither is used alone. DLR is the sharper limitation—admitted on its target family, where it responds perfectly, but with an inverted rotational response (−0.80), so it is read only as a comparative diagnostic alongside DAR. Reported alongside these is the scope boundary of Sec. 5.1: looping content sustains both persistence and magnitude, so SNF-Bench cannot distinguish genuine progression from repetition, and does not claim to.

6. Public-Model Audit

We apply SNF-Bench to publicly released long-horizon generators. The goal is not a universal leaderboard but

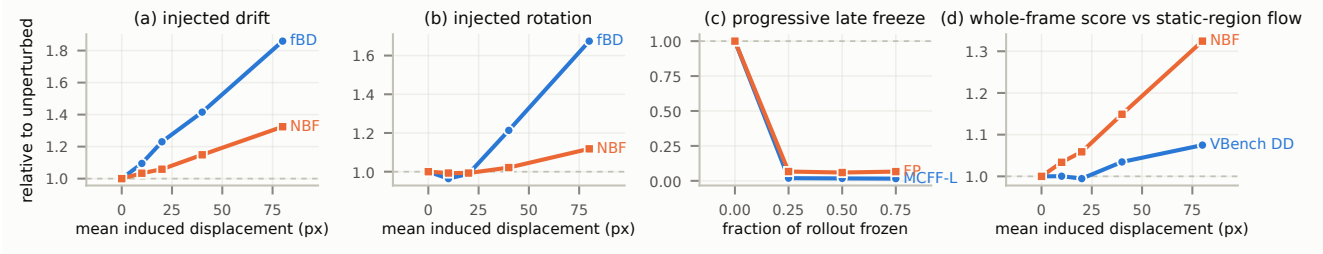


Figure 2. Mechanistic validation. Each panel injects a corruption of known kind and severity into real fixed-camera clips and plots each factor relative to its unperturbed value. Panel (c) shows the progressive late-freeze intervention the persistence factors are admitted on, so figure and criterion are one experiment: freezing the late window collapses MCFF-L and FP to a few percent of their unperturbed value. Panel (d) puts VBench Dynamic Degree and static-region NBF through the same synthetic translation. Static-region flow rises strongly while the whole-frame score changes little and slightly upward, illustrating that a generic apparent-motion score does not identify where motion occurs. The perturbation is synthetic, so the comparison implicates no published method.

a determination of how checkpoints occupy the static-fidelity/flow-persistence space under a single recorded T2V setting.

Systems and setting. The audit covers 7 publicly released autoregressive or streaming generators with accessible checkpoints and reproducible inference code, frozen as a set before the sweep. Every T2V checkpoint runs through one common inference configuration rather than each method’s released defaults: a four-step denoising schedule, guidance 5.0, six frames per block and a fixed seed, with only the checkpoint and horizon varying. All checkpoints are evaluated at the same output resolution, frame rate, target horizons, and wrapper settings. Accordingly, the results isolate checkpoint behavior under this common deployment rather than reconstructing each method’s published inference procedure. A checkpoint distilled for a different step budget is therefore not observed at its own operating point. The I2V track separately distinguishes recorded native outputs from checkpoints run through its common long-horizon wrapper (Sec. 7.3).

Controls and statistics. Within each item every system receives identical conditioning, and the seed set is fixed before the sweep. Seed control guarantees reproducibility within a system, but because implementations consume randomness differently, identical seeds do not induce matched noise draws across systems; cross-system comparison therefore rests on the paired-item design. Videos are retained in their recorded form and the normalization of Sec. 4 applied unchanged. Scores are computed per item and averaged over items; the sensitivity of every conclusion to category macro-averaging is reported in the supplementary material. Uncertainty is estimated by per-

centile bootstrap [2] over items, and we report 95% marginal intervals. These intervals summarize per-checkpoint uncertainty; all headline separation claims use paired bootstrap distributions of per-prompt differences, reported in Supplementary Table S3. Operational limits—unsupported horizons, failed generations, memory exhaustion, invalid outputs—are reported as rates against the frozen manifest and never removed after inspection. To test whether the decomposition changes interpretation, the same outputs are additionally scored with established whole-frame metrics and the induced orderings compared. The focus is interpretation change, not a claim that existing metrics are wrong.

7. Results and Interpretation Changes

Unless stated otherwise, T2V comparisons refer to checkpoints under the common matched setting. I2V native-output and wrapper-based results are identified separately and are never ranked across settings.

7.1. Matched T2V Checkpoint Audit

Table 2 reports the text-conditioned (T2V) track at 60s, the tier whose sample supports method-level separation, and at 120s as a supporting tier. The 120s panel carries 6 prompts and is read as corroboration of direction rather than as an independent separation; it draws a different prompt set from the 60s panel, so the two panels are not compared column-by-column. The 5 and 240s tiers, an initialization check and a diagnostic extreme respectively, are in the supplementary material with all four horizons.

Motion persistence. Dynamic-region behavior is read at the joint point (MCFF(L), FP), never FP alone:

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 23$)				
CausVid	8.08 $\pm$ 3.15	9.27 $\pm$ 4.14	2.87 $\rightarrow$ 1.96	0.67 $\pm$ 0.18
Self-Forcing	11.4 $\pm$ 3.63	12.7 $\pm$ 3.97	10.4 $\rightarrow$ 2.54	0.44 $\pm$ 0.22
Infinite-Forcing	7.75 $\pm$ 3.84	3.65 $\pm$ 0.88	1.17 $\rightarrow$ 0.29	0.63 $\pm$ 0.22
Rolling-Forcing	16.8 $\pm$ 3.69	11.2 $\pm$ 2.70	3.18 $\rightarrow$ 1.57	0.75 $\pm$ 0.29
Reward-Forcing	14.8 $\pm$ 3.71	8.51 $\pm$ 2.79	3.81 $\rightarrow$ 0.80	0.49 $\pm$ 0.18
LongLive	15.9 $\pm$ 2.89	13.9 $\pm$ 4.76	6.46 $\rightarrow$ 2.41	0.75 $\pm$ 0.28
Causal-Forcing	22.4 $\pm$ 2.26	87.6 $\pm$ 18.6	9.84 $\rightarrow$ 7.21	0.83 $\pm$ 0.31
120s horizon ($n = 6$)				
CausVid	6.15	3.42	0.36 $\rightarrow$ 0.52	1.06
Self-Forcing	21.2	6.80	4.85 $\rightarrow$ 0.82	0.85
Infinite-Forcing	16.1	2.60	0.50 $\rightarrow$ 0.32	0.64
Rolling-Forcing	23.8	10.6	3.33 $\rightarrow$ 0.77	0.38
Reward-Forcing	20.0	11.6	2.05 $\rightarrow$ 0.69	0.39
LongLive	22.5	9.66	5.10 $\rightarrow$ 1.19	0.81
Causal-Forcing	21.8	130	10.5 $\rightarrow$ 10.7	0.90

Table 2. Matched T2V checkpoint audit. Public checkpoints are evaluated under one common four-step, guidance-5.0, six-frames-per-block, fixed-seed setting; results characterize this wrapper, not released inference procedures. n is common within each horizon. Entries are prompt means $\pm$ half a percentile-bootstrap 95% interval (10k resamples, fixed seed); $n < 10$ panels show means only. MCFF-E $\rightarrow$ L contextualizes the FP ratio. Marginal intervals are descriptive; separation claims use paired per-prompt bootstrap differences (Table S3). DLR/DAR and aggregation sensitivity are in the supplement.

Method	fBD↓	NBF↓	MCFF E→L	FP↑
60s horizon ($n = 30$)				
Causal-Forcing++	9.18	5.97	0.85 $\rightarrow$ 0.60	0.86
(2-step) [†]				
Causal-Forcing++	5.34	7.05	0.86 $\rightarrow$ 0.57	0.77
(1-step) [†]				
Causal-Forcing	21.1	36.4	2.80 $\rightarrow$ 2.69	0.89
(frame-wise) [†]				
Causal-Forcing++	9.94	6.91	1.30 $\rightarrow$ 0.78	1.02
(2-step, frame-wise)				
Self-Forcing [†]	17.3	5.52	1.50 $\rightarrow$ 0.69	0.89
CausVid [†]	19.9	3.31	0.58 $\rightarrow$ 0.41	0.81
Causal-Forcing++	10.4	5.25	0.53 $\rightarrow$ 0.52	1.01
(1-step, frame-wise)				

Table 3. I2V released pipelines and deployment sensitivity. Unmarked rows are recorded released-pipeline outputs; [†] rows use the common long-horizon wrapper and are not compared with them. n is common within each horizon. Entries are prompt means; intervals are in the supplement. MCFF-E $\rightarrow$ L contextualizes the FP ratio. No claim compares settings. DLR/DAR and aggregation sensitivity are in the supplement.

trivial-motion behavior without collapsing the axes into a scalar score; the supplementary material plots this. The extremes are informative in opposite directions. The Infinite-Forcing checkpoint under the common setting [9] has the lowest normalized background flow: a paired bootstrap over per-prompt differences separates it from all six other checkpoints, every interval excluding zero (supplementary). It also shows the least surviving late motion. The Causal-Forcing checkpoint under the common setting [27] occupies the opposite corner, with substantially higher background flow.

7.2. Whole-Frame Motion and Static-Region Flow Yield Different Interpretations

The effect is systematic rather than anecdotal. Correlating each generic whole-frame metric with each SNF-Bench factor over checkpoint-level means (supplementary), Dynamic Degree tracks static-region flow at $\rho = +0.93$: under the common setting, checkpoints with more apparent motion also tend to have more background motion. The Causal-Forcing checkpoint ranks 1 of 7 on Dynamic Degree and 7 on NBF; the Infinite-Forcing checkpoint mirrors it, ranking 1 on static fidelity and 7 on apparent motion. Read with its magnitudes, the Causal-Forcing checkpoint’s DLR of 0.94 indicates static-region motion energy almost equal to the region intended to move, while its DAR of 0.40 indicates that a rigid global model accounts for only part of that motion. What the residual contains is not identified here—it may include non-rigid instability, intended local motion, partition contamination or flow error—and these measurements do not separate them.

7.3. Image-Conditioned (I2V) Track: Released Pipelines and Deployment Sensitivity

At 60s, Table 3 keeps two recorded released-pipeline outputs separate from five dagger-marked deployment-sensitivity rows run through the common long-horizon wrapper. It is not a seven-system audit, and the settings are never ranked against one another. The panel shows only that deployment can materially change checkpoint scores; why it does so is outside what a benchmark establishes. Unlike T2V, every I2V system starts from the same source image, so static support is given rather than differently imagined. Complete horizons, including bidirectional references at 5s, are in the supplement.

8. Limitations and Conclusion

SNF-Bench is intentionally narrow: long-horizon fixed-camera generation, where static support and dynamic

581 sustained flow needs non-trivial late motion and high
582 retention. MCFF-E beside MCFF-L exposes the de-
583 generate case, since negligible motion throughout can-
584 not reach a high FP without a small MCFF-E.

585 Operating regimes. Read jointly, the static-fidelity
586 and persistence columns separate low-drift/low-
587 motion, high-motion/high-drift and low-drift/non-

flow separate. It measures neither physical realism nor semantic correctness, cannot distinguish progression from repetition, and rests on an automatic partition whose limits are stated in Sec. 3.2 and bounded by the erosion–dilation study. Conclusions unstable under that perturbation are never headline claims.

SNF-Bench shows that whole-frame motion and region-resolved static fidelity can produce materially different interpretations of the same fixed-camera long-horizon outputs. By reporting static fidelity and flow persistence separately, with drift-related quantities used as context rather than as a single ranking score, it complements broad video-generation benchmarks without claiming physical or semantic realism. The released specification and matched-checkpoint audit provide a reproducible basis for future evaluation, while independent partition validation and broader non-water coverage remain important next steps.

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